# BUSINESS FINANCE

*Formulae Sheet — Quick Reference*

B.B.A. Programme — CSJMU, Kanpur

## 1. Time Value of Money

| Item | Formula | Key Terms |
|---|---|---|
| Future/Terminal Value (single sum) | $FV = PV\,(1 + i)^n$ | PV=present value, i=rate, n=periods |
| Compounding (m times a year) | $FV = PV\,(1 + i/m)^{(n\times m)}$ | m = compounding frequency per year |
| Present Value (single sum) | $PV = FV / (1 + i)^n$ | |
| Future Value of an Annuity | $FVA = A\,[\,((1+i)^n - 1) / i\,]$ | A = periodic instalment |
| Present Value of an Annuity | $PVA = A\,[\,(1 - (1+i)^{-n}) / i\,]$ | |
| Present Value of Perpetuity | $PV = A / i$ | A = constant periodic amount |
| Annuity Due (FV / PV) | $Value(due) = Value(ordinary) \times (1+i)$ | Payments at start of period |
| Loan Amortisation Instalment | $A = Loan\ Amount / PVIFA(i, n)$ | PVIFA = $[1-(1+i)^{-n}]/i$ |
| Interest for a period (loan) | $Interest = Opening\ Balance \times i$ | Principal repaid = Instalment − Interest |

## 2. Valuation of Bonds

| Item | Formula | Key Terms |
|---|---|---|
| Bond Value (annual interest, redeemable) | $V0 = I \times PVIFA(kd,n) + F \times PVIF(kd,n)$ | I=annual interest, F=face/redemption value, kd=required yield |
| Bond Value (semi-annual interest) | $V0 = (I/2)\times PVIFA(kd/2, 2n) + F\times PVIF(kd/2, 2n)$ | |
| Perpetual (Irredeemable) Bond | $V0 = I / kd$ | No maturity |
| Zero-Coupon Bond | $V0 = F / (1+kd)^n$ | No periodic interest |

*Bond price rule: Coupon = Yield → sells at Par; Coupon > Yield → Premium; Coupon < Yield → Discount.*

## 3. Valuation of Stocks & Return

| Item | Formula | Key Terms |
|---|---|---|
| Share Value — single period | $P0 = D1/(1+ke) + P1/(1+ke)$ | D1=dividend, P1=year-end price, ke=required return |
| Share Value — constant (no-growth) | $P0 = D / ke$ | Perpetuity of dividends |

| | | |
|---|---|---|
| dividend | | |
| Rate of Return from a Stock | ***ke = [D1 + (P1 – P0)] / P0*** | Dividend yield + Capital gains yield |
| Gordon Growth (Constant Dividend Growth) Model | ***P0 = D1 / (ke – g)*** | D1 = D0(1+g); g = constant growth rate; ke > g |
| Cost of Equity from Gordon Model | ***ke = (D1 / P0) + g*** | |

## 4. Capital Budgeting

| Item | Formula | Key Terms |
|---|---|---|
| Payback Period (equal cash inflows) | ***Payback = Initial Investment / Annual Cash Inflow*** | |
| Payback Period (unequal cash inflows) | ***Payback = E + (B / C)*** | E=yrs before recovery, B=balance to recover, C=inflow in recovery year |
| Accounting Rate of Return (ARR) | ***ARR = (Avg. Annual PAT / Avg. Investment) × 100*** | Avg. Investment=(Initial−Salvage)/ 2 + Salvage |
| Net Present Value (NPV) | ***NPV = Σ [CFt /(1+k)$^t$] − Initial Investment*** | CFt=cash inflow at time t, k=cost of capital |
| Internal Rate of Return (definition) | ***Σ [CFt /(1+r)$^t$] = Initial Investment*** | Solve for r (IRR); NPV=0 at r |
| IRR by Interpolation | ***IRR = LR + [NPVLR/(NPVLR−NPVHR)] × (HR−LR)*** | LR/HR = lower/higher trial discount rates |

*Decision Rules: Accept if Payback < standard period | ARR > cut-off rate | NPV > 0 | IRR > cost of capital.*

## 5. Leverage & Financing Decisions

| Item | Formula | Key Terms |
|---|---|---|
| Degree of Operating Leverage (DOL) | ***DOL = Contribution / EBIT*** | Contribution = Sales − Variable Cost |
| Degree of Financial Leverage (DFL) | ***DFL = EBIT / (EBIT − Interest)*** | |
| Degree of Combined Leverage (DCL) | ***DCL = DOL × DFL = Contribution/(EBIT−Interest)*** | |
| Earnings per Share (EPS) | ***EPS = (EBIT − I)(1 − t) / No. of Equity Shares*** | t = tax rate, I = interest |
| EPS with Preference Dividend | ***EPS = [(EBIT−I)(1−t) − Pref. Dividend] / No. of Eq. Shares*** | |
| Indifference Point (EBIT-EPS) | ***(EBIT−I1)(1−t)/N1 = (EBIT−I2)(1−t)/N2*** | Plans 1 & 2: interest I, shares N |
| Interest Coverage Ratio | ***= EBIT / Interest*** | Debt-servicing cushion |
| Debt Service Coverage Ratio (DSCR) | ***= (PAT + Interest + Depreciation) / (Interest + Principal Instalment)*** | |

*Pecking Order of Financing: Internal funds (retained earnings) → Debt → Hybrid securities → New Equity (last resort).*